PROBLEM STATMENT

E-learning platforms focus on content delivery and completion metrics but fail to capture true understanding, engagement depth, and verifiable learning outcomes.

SOLUTION

KNOX

TRACK

Web & App Dev

INSTITUTE

Vishwakarma Institute of Technology, Pune

TEAM NAME

The Solver Squad

TEAM MEMBERS

1.Shantanu Gudmewar
2.Ruturaj Ambure
3.Atharv Bhavsar
4.Hemant Bhoyar

Problem Statement

Passive Consumption

- Platforms track **watching**, not **understanding**
- Students click **"Next"** without thinking
- Completion ≠ Comprehension
- **220M+** learners, yet **55%+** drop out / low retention

Fake Personalization

- Same **content**, same **path** for everyone
- No **adaptation** to student understanding
- Platforms serve **100M+** learners with one fixed flow
- System does not know the **learner**

Certificates Prove Nothing

- **Completion + retention**-based validation
- **Micro-checks** over time **(Day 1, 7, 30, 90)**
- **Skill-based profile** instead of static certificate
- Follow-up questions to verify **real understanding**

Fragmented Ecosystem

- **Learning + Practice + AI** Tutor in one system
- **Context-aware** problem solving (no platform switching)
- Socratic AI (guides, not answers)
- One **unified student profile** (tracks everything)

Watched ≠ Learned → Recommended ≠ Personalized → Certified ≠ Retained → Answered ≠ Understood

Existing solutions & their limitations

Current Lending System

- **Video-based** learning (MOOCs)
- **Linear course structure** for all learners
- Basic quizzes after content
- **Coding platforms** separated from learning
- **Certificates** based on **course completion**
- **No tracking** of actual understanding

Key Gaps / Limitations

- **Passive learning** → students just watch
- No real-time cognitive check (**thinking not verified**)
- **Same path** for every learner (no adaptation)
- **Practice disconnected** from concepts
- One-time evaluation → **no retention tracking**
- Certificates do not **reflect** real knowledge

OUR Solution

- Forces **active thinking** via auto-pause + questions
- **Tracks learner state** in real-time (confused / strong / struggling)
- Dynamic learning path based on **mastery score**
- Integrated **learning + coding** practice system
- **Continuous evaluation** (Day 1, 7, 30, 90 retention checks)
- Proof-based **dynamic certification** (not static)

Today's learning systems measure activity, not understanding.
KNOX transforms learning into an active, adaptive, and continuously proven process.

Introducing - KNOX

K
Knowledge Mapping

Structured, interconnected concept graph

N
Neural State Tracking

Real-time cognitive understanding detection

O
Outcome Validation

Proof-based learning verification

X
Experience Adaptation

Dynamic, personalized learning flow

From watching content to proving knowledge — KNOX makes learning real.

Unique Selling Propositions

Cognitive Engagement

- System **forces thinking** before progressing
- **Predict → Explain → Continue**
- **Feynman-based validation** (explain in own words)
- **Detects confusion** (rewind, pause, idle)

Real Personalization

- **Tracks understanding** after every concept
- Adjusts **difficulty in real-time** (easy / medium / hard)
- **Dynamically changes** learning path
- Course **adapts** to student, not vice versa

Proof-Based Intelligence Layer

- **Dynamic certification** (based on real performance)
- Knowledge graph detects **weak areas**
- **AI mentor** guides learning (not direct answers)
- Proves and improves **real understanding**

Unified Learning System

- **Learning + Practice + AI** Tutor in one system
- Context-aware problem solving (**no platform switching**)
- Socratic AI (guides, not answers)
- One **unified student profile** (tracks everything)

AI-Based Liveness Detection

- Blink-based human **verification**
- Real-time face & head **tracking**
- Continuous presence monitoring
- Anti-spoofing via multi-signal validation

Implementation Plan

1 Skeleton Flow (Phase - 1)

- One hardcoded topic with static content
- Video player with auto-pause at 2 minutes
- Interrupt prompt with minimum 20 word response
- Basic score display (correct +10, wrong -5)

2 Adaptive Engine(Phase - 2)

- Mastery score stored in MongoDB per topic
- Micro-check question after every concept
- Score < 40 → easy, 40–70 → medium, > 70 → advanced

3 Practice Engine(Phase - 3)

- Monaco Editor embedded in platform
- Judge0 API runs student code against test cases
- Pass → mastery increases, move to harder problem
- Fail → hint appears, easier problem served

4 Feynman Gate + AI Tutor(Phase - 4)

- Student explains concept in own words
- Basic AI evaluation (simple prompt check)
- One follow-up question (not full chain)
- Basic Socratic behavior (1–2 guided questions only)

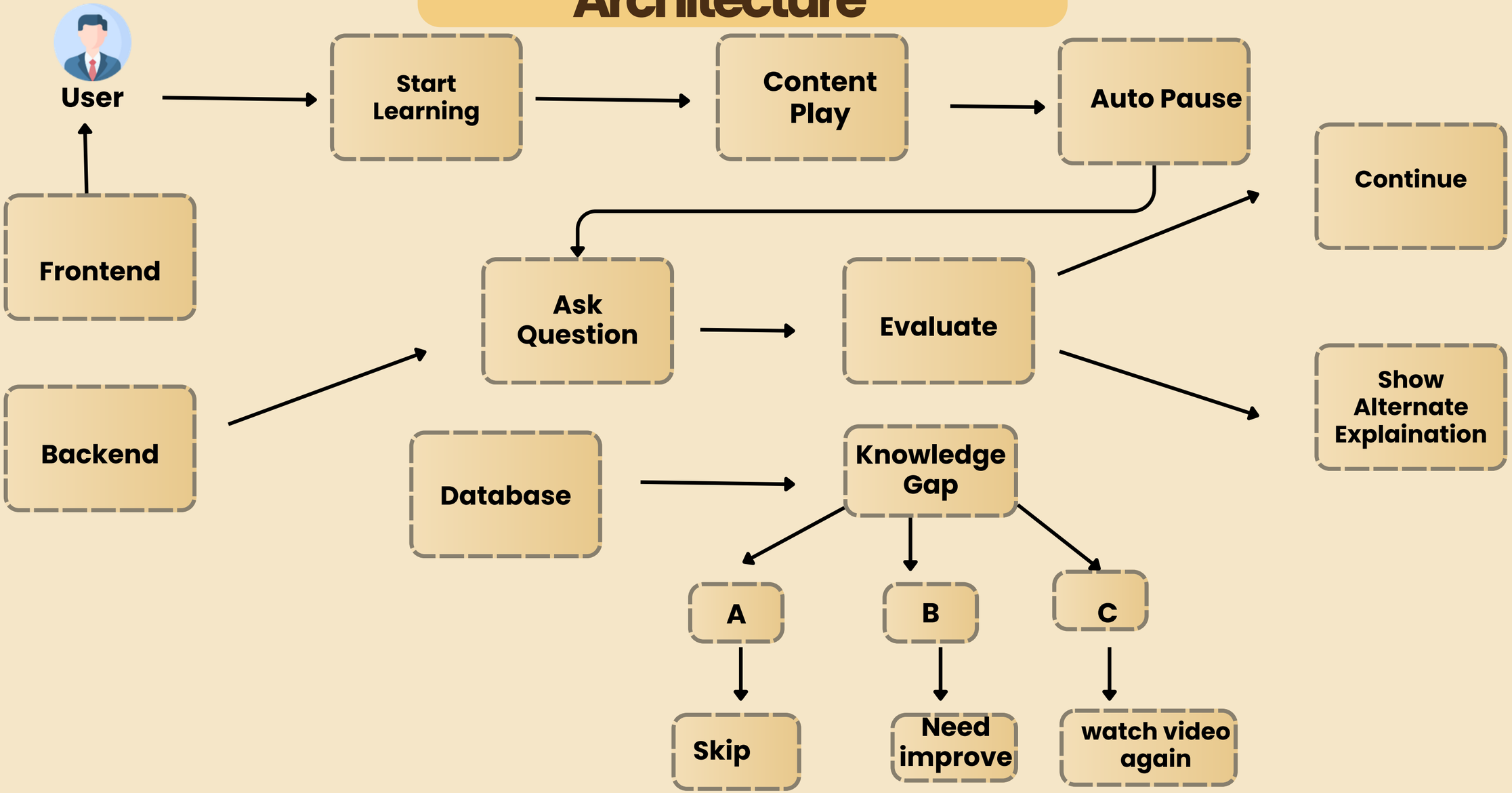
5 Verification System(Phase - 5)

- Course completion generates certificate (UI only)
- Show retention timeline (static/demo data)
- Certificate states:
- Completed
- Verified (UI simulation)
- Topic-wise skill breakdown shown

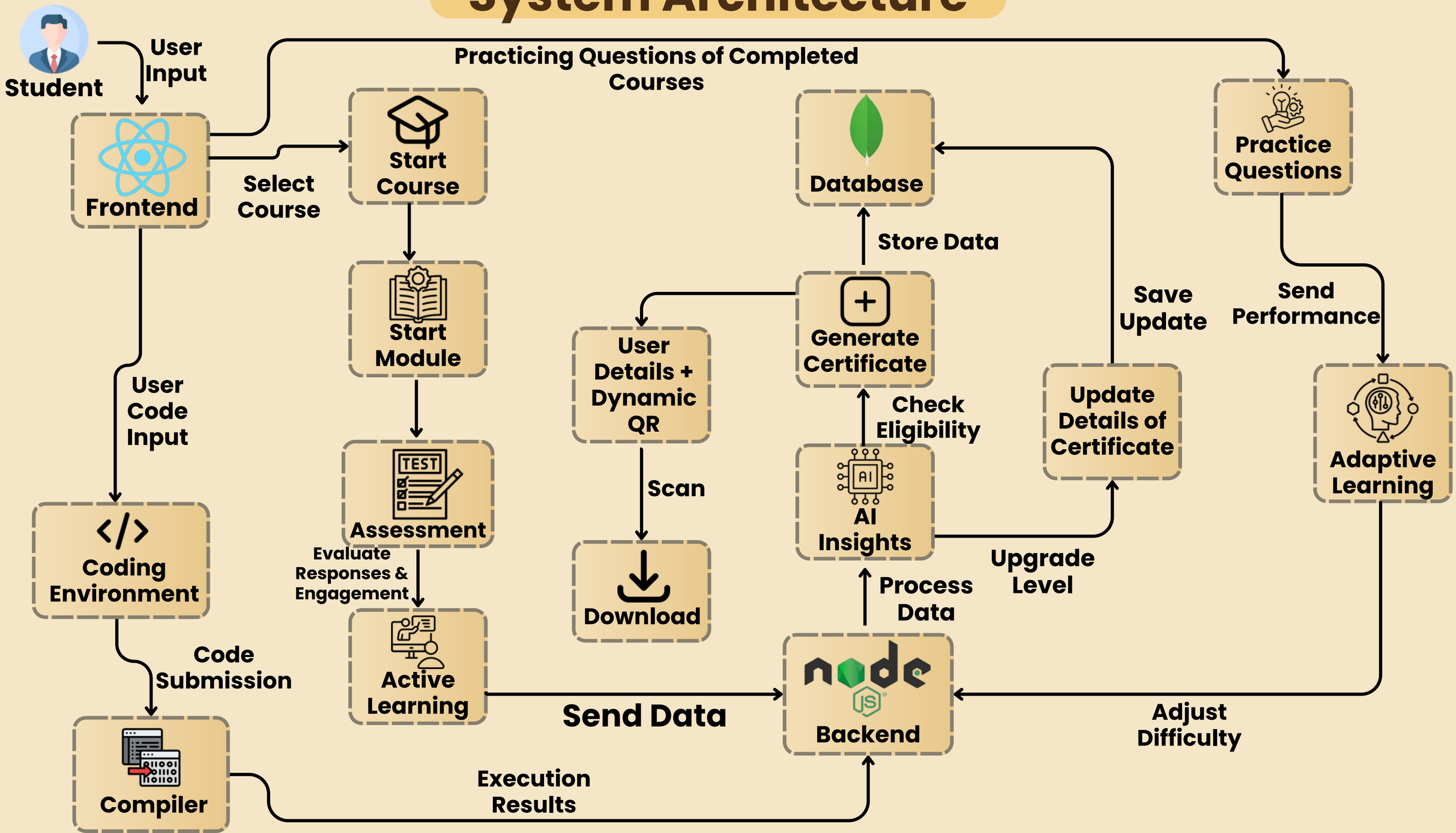
6 Dashboard + Polish(Phase - 6)

- Topic-wise mastery bars
- Strong vs weak topics display
- Smooth full demo flow
- Clean UI + loading states

Architecture



System Architecture



Technology Stack

Frontend

React

MONACO

tailwindcss

Backend

node JS

Express JS

External APIs

cohere

Database

mongoDB

redis

Deployment

Vercel

Render

Feasibility & Viability

Challenges

- Learning platforms cannot verify real understanding
- Difficult to track cognitive state in real-time
- Students tend to passively consume content
- No system ensures long-term retention

Mitigations

- Auto-pause + micro-questions enforce thinking
- AI tracks learner state in real-time
- Adaptive paths based on mastery
- Re-testing (Day 1, 7, 30, 90) ensures retention

Viability & Feasibility

- Built on scalable modern stack (React, Node.js, AI APIs)
- Works with existing content + coding platforms
- No hardware dependency → fully software-based system
- Easily deployable across colleges, ed-tech, and enterprises
- Ready to scale, not just a prototype

KNOX does not measure how much you watched — it proves how much you actually understand and retain.

Impact

Impact

- Converts watching → thinking → proving
- Ensures understanding, not just completion
- Enables real-time personalization at scale
- Builds measurable learning intelligence

Benefits

- Higher retention through spaced repetition system
- Stronger skills via learning + practice integration
- Transparent proof via dynamic certification
- Early detection of weak concepts and gaps

Cognitive Intelligence

Adaptive Learning Engine

Continuous Verification

Business Model

Freemium Model

- Free access → 80–90% users onboard
- ₹199–₹499/month for AI tutor & adaptive learning
- Target: 5–10% conversion → high recurring revenue

B2B Institutional Sales

- Colleges/companies pay ₹50K–₹5L/year per license
- Includes analytics, dashboards & skill reports
- Focus: long-term contracts + bulk users (1000+)

Certification Revenue

- Basic certificate → Free
- Verified certificate → ₹299–₹999
- Builds trust for employers → higher demand

Scalable AI platform for faster, safer, and smarter lending decisions.

Thank you!

Presented By The Solver Squad